



Print my heart

Surgeons in Japan use 3-D printing to help trim a liver for transplanting.
<http://dgit.in/10Wb93H>

30 day trip

University of Washington scientists are building nuclear-fusion powered rockets allowing a trip to and fro Mars, in 30 days. <http://dgit.in/10Jl5x5>

Racing Ahead to a Distant Reality

Can the new fusion driven rockets change the way we see space travel?

Cyril Vattelli

readersletters@thinkdigit.com

“Honey I’ll be back in two months” “Another business trip to Zurich?” “No baby. Mars” Yes. Travel to Mars in two months is nearing a reality. Scientists at the University of Washington have a proposal which can make travel to Mars within thirty to ninety days possible. The new nuclear Fusion powered rockets are the key. Take a break from this new system to understand why it is so important.

NASA has accomplished six successful landings on Mars: Viking I and II, Pathfinder, MER, Spirit and Opportunity, and Phoenix. But these were all unmanned missions with lighter spacecrafts due to the absence of astronauts, supplies, and fuel. Another big factor is that no one is going to really shed a tear if a robot gets stuck on Mars. So clearly, there is something stopping humans from travelling to Mars. But why is travel to mars so difficult? Mars is 150 times further away than the moon is, and that is just one thing. The immense psychological implications of being stuck in a closed environment with other people is enough to drive someone to



Biosphere 2: The return of biosphere

depression. To give an idea of how damaging this is, in September 1991, eight people, dubbed ‘biospherians’ were sealed off inside a 3.14 acre glass enclosure located in Arizona called Biosphere 2, Biosphere 1 being where we live in right now. The idea was for humans to develop and live in a self sustaining environment and was considered a parallel to living in outer space. It started

off well enough with the biospherians recycling their wastes and producing their own food while at the same time conducting ecological research. By April 1992, things had changed. Two distinct groups had emerged which had opposing views on how to live. There was some scarcity of food so the animals were slaughtered and eaten and some even ate the seeds meant for growing crops. Eventually, people even started smuggling food from the outside. A second mission was initiated in March 1994 but again failed in six months. Now consider the same scenario in space. There is no ‘fall-back’ plan of smuggling anything from the outside. Most importantly, the people in Biosphere 2 knew that they are still on Earth, albeit in a bubble, but in space there is no backing out when one can’t take any more. Feelings of isolation and boredom, the knowledge that Earth is so distant, and long periods of confinement are some of



They clearly put all their money in research, and not in their 3-D models



Will it look less grand once we are able to get there easily?